AI Spacecraft Anomaly Detection System

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- ITAI 2372
- Conceptual Design Project
- Using AI to detect spacecraft failures early

Problem

- Spacecraft systems generate large amounts of real-time data
- Engineers cannot monitor everything at once
- Small issues can be missed
- Missed issues can become serious failures

Proposed Solution

- AI system monitors spacecraft data
- Learns normal behavior
- Detects unusual patterns
- Sends alerts when something is wrong

System Components

- Data Collection (sensors)
- Data Processing
- AI Model (anomaly detection)
- Alert System
- Engineer dashboard

How the System Works

- Collect sensor data
- Process and clean data
- AI analyzes patterns
- Detect anomalies
- Send alerts

AI Techniques

- Anomaly Detection
- Isolation Forest (detects outliers)
- LSTM (handles time-based data)

System Flow

- Sensors → Data Processing → AI Model → Alert → Engineer

Testing Plan

- Test with normal data (no alerts expected)
- Test with abnormal data (alerts expected)
- Measure accuracy
- Check false positives and false negatives

Benefits

- Detects problems early
- Improves safety
- Reduces human workload
- Faster decision-making

Limitations

- May give false alerts
- Needs high-quality data
- May miss rare issues

Conclusion

- AI improves spacecraft monitoring
- Helps detect issues early
- Reduces mission risk
- Supports engineers